Projected Gradient Descent for LDPC Decoding: From a Smooth Parity Penalty to Explicit Constraints

BP–GD Mechanism Study

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Abstract

This paper explains a persistent performance gap between native exact sum-product belief propagation (BP) and a family of projected-gradient decoders for a length-10,200 $(3, 6)$-regular low-density parity-check code with 5,100 parity checks. On registered paired cases, native projected gradient descent recovered $112/200$ transmitted words at 6 dB and $0/200$ at 1.5 dB after 30 updates, whereas exact BP recovered $200/200$ and $199/200$, respectively. A later compound-box projected-gradient construction reached $200/200$ after 90 updates on a separate registered 6-dB population, but it has no completed 1.5-dB transfer. A related registered scalar control remained at $0/200$ at 1.5 dB after both 30 and 90 updates.

We show analytically how the smooth product parity term can assign zero cost to both a correct local sign pattern and a pattern with two wrong signs, while a continuous correction between them incurs positive cost. A constructed two-variable example then proves that an objective with the implemented term shapes can possess a strict wrong local minimum on the box even when the desired word has much lower objective value. Exact projected stationarity precludes escape by additional ordinary local descent, but it does not exclude momentum, nonmonotone jumps, state-dependent matrices, different initializations, different objectives, or expanded-variable methods. We therefore derive a differentiable convex parity cost based on distance to one parity polytope per check and explain why finite quadratic penalties do not generally impose exact feasibility. On an exposed 200-case population at 1.5 dB, ordinary box-projected descent of one frozen instance of this cost recovered $0/200$ words at every checkpoint through update 6000 and produced no parity-valid readout; all 200 terminals remained above the registered projected-gradient-mapping tolerance and are therefore optimization-unresolved, not identified minimizers. A replica variable for each check, together with multipliers, can enforce consistency without recreating the same product hill, although the loopy intersection remains the fundamental-polytope relaxation and can have fractional vertices. The new count is an exposed-population mechanism diagnostic, not a two-SNR confirmation or comparison with BP.

1 Introduction

Projected-gradient decoding is attractive because its operations are differentiable, parallelizable, and compatible with parameter training. Prior work has formulated low-density parity-check (LDPC) decoding as nonconvex projected-gradient optimization with smooth parity penalties and has used restarts or learned parameters to address local minima [1]. Linear-programming (LP) decoding takes another route: it minimizes a linear channel cost over a relaxation derived from the parity checks [2]. Efficient decomposition methods introduce replica variables and Euclidean projection onto each parity polytope [3].

The present study began with a narrower engineering question: why does a specific projected-gradient construction work well at 6 dB yet remain far behind exact sum-product BP at 1.5 dB? The campaign progressively separated state, direction, step rule, check assignment, and update budget. It found that direct

application of an exact recurrence can recover words where tested true-gradient arms do not, but direct recurrence application is a fixed-point update rather than descent of the declared objective.

The resulting claim is deliberately bounded. We establish that the implemented objective *can* contain wrong local minima on the box and that an exact projected zero cannot be escaped by further ordinary projected descent. We do not establish that every 1.5-dB failure has been registered as an exact local minimum, and we do not claim that every method involving the partial derivatives of the objective function must fail.

The contributions are fourfold:

1. We organize the controlled evidence without combining distinct case populations or treating a direction fit as a decoder result.
2. We derive the partial derivatives at a hard vertex and isolate what happens when one check contains two wrong signs.
3. We give a strict constructed local minimum on the box and state precisely what exact projected stationarity closes.
4. We derive a differentiable convex parity cost, prove its exactness limitation, report an exposed-population diagnostic, and motivate one parity-polytope constraint per check with multipliers as a scientifically distinct successor formulation.

2 Controlled Evidence and Method Boundary

All comparisons in the initial paired experiment held fixed the parity-check matrix, transmitted word, noise realization, and vector of channel log-likelihood ratios (LLRs). A positive LLR favors bit zero. Recovery means that the hard-decision estimate equals the transmitted word; parity validity alone is not counted as recovery.

Here *native exact sum-product BP* means that the implementation evaluates the declared sum-product recurrence without approximation. It does not mean exact maximum a posteriori or maximum-likelihood inference on the loopy Tanner graph.

Exact sum-product BP retains directed variable-to-check and check-to-variable messages and performs recipient exclusion. The scalar projected-gradient control retains one scalar $x_i \in [-1, 1]$ for bit i, calculates the partial derivatives of its objective function, applies its declared direction and step rule, and projects componentwise onto the box. These methods traverse the same Tanner graph but do not retain or update the same quantities.

Table 1: Registered recovery evidence. Every entry is the number of transmitted words recovered out of 200 cases. Rows with different case populations are not a single cross-SNR decoder evaluation.

Experiment and population	SNR and budget	Projected-gradient or recurrence result	Exact BP or scope boundary
Native composition, frames 60025–60224	6 dB, 30 updates	Native projected GD: 112	Exact BP: 200
Native composition, frames 60225–60424	1.5 dB, 30 updates	Native projected GD: 0	Exact BP: 199
Selected compound-box, frames 50025–50224	6 dB, 30 / 90 updates	177 / 200	No completed exact 1.5-dB transfer
Native compound budget diagnostic, frames 61745–61944	6 dB, 30 / 90 updates	113 / 193	Separate population and construction
Native compound budget diagnostic, frames 61969–62168	1.5 dB, 30 / 90 updates	0 / 0	Closes this frozen budget extension only
Convex parity-distance diagnostic, frames 61969–62168	1.5 dB, 30 / 90 / 300 / 1000 / 3000 / 6000 updates	0 / 0 / 0 / 0 / 0 / 0	Exposed population; all 200 cap terminals unresolved; BP absent from candidate packet/source
State–direction–step factorial, frames 70921–71120	1.5 dB, 30 updates	Free state + recurrence displacement + unit application: 196; every tested true-gradient cell: 0	Successful cell is a fixed-point update, not a gradient step

The strongest retained gradient-based recovery result is the selected compound-box construction at 6 dB: $200/200$ transmitted words after 90 updates. It is not a cross-SNR result approaching BP. Conversely, the $0/200$ registered result at 1.5 dB through 90 updates closes the claim that the frozen scalar control merely needed that larger cap; it is not a theorem about every step sequence or every gradient-based decoder.

In the state–direction–step factorial, direct unit application of the recurrence minus the current state recovered $196/200$ words at 1.5 dB, whereas all tested true-gradient cells recovered $0/200$. This establishes an update-operator boundary. It does not establish that BP succeeds only by increasing the objective, nor does endpoint nonincrease certify monotonicity inside a finite update.

3 The Implemented Smooth Objective

Let $x \in [-1, 1]^n$ denote the mean coordinate on which the compound objective depends. The implemented objective is

$$J_C(x) = -\frac{L^{\mathsf{T}}x}{2n} + \frac{w_p}{n}\sum_c \left(1 - \prod_{i\in c} x_i\right)^2 + \frac{w_b}{n}\sum_i (1 - x_i^2)^2, \tag{1}$$

where $w_p = 2$ and $w_b = 0.2$. The production evaluator clips each LLR to $[-20, 20]$. Define

$$p_c = \prod_{j\in c} x_j, \qquad q_{ci} = \prod_{\substack{j\in c \\ j\neq i}} x_j. \tag{2}$$

The exact partial derivative is

$$\partial_i J_C(x) = -\frac{L_i}{2n} - \frac{2w_p}{n}\sum_{c\ni i}(1-p_c)q_{ci} + \frac{4w_b}{n}x_i(x_i^2-1). \tag{3}$$

The implementation may store directed-edge coordinates z, but the objective depends only on their incident-edge means $x = Cz$. Thus $J_C(z) = J_C(Cz)$, and directions in $\ker C$ are exactly flat. Claims about a well or barrier must therefore be made in the mean coordinate, not by treating the full edge state as an isolated minimum.

3.1 Partial derivatives at a hard vertex

Lemma 1 (Partial derivatives at a hard vertex)**.** *Let $s \in \{-1,+1\}^n$, and let k_i be the number of violated checks incident on bit i. Then*

$$\partial_i J_C(s) = -\frac{L_i}{2n} + \frac{4w_p}{n}k_i s_i. \tag{4}$$

Proof. At a satisfied check, $p_c = 1$, so its contribution to Equation (3) is zero. At a violated check, $p_c = -1$ and $q_{ci} = -s_i$, so its contribution is $4w_p s_i/n$. The final term of Equation (3) is zero at a hard vertex. Summing the k_i violated-check contributions gives Equation (4). □

One violated check therefore applies the same parity magnitude to all six of its incident coordinates at a hard vertex. The complete partial derivative can still distinguish those coordinates through the channel LLRs, their total numbers k_i of violated incident checks, overlapping checks, and, away from the vertex, the partner products and the final term of Equation (3). Exact sum-product BP instead forms a check-to-variable message from the other five incoming variable-to-check messages. Recipient exclusion is the structural contrast; Equation (4) alone is not a decoder-performance theorem.

The registered initialization is $x_i^{(0)} = \tanh(L_i/2)$ rather than a hard vertex. The lemma consequently explains a local mechanism, not the exact initial trajectory.

4 How Two Wrong Signs Create an Uphill Route

At a hard-sign vector, a check containing zero wrong signs has product $+1$, a check containing one wrong sign has product -1, and a check containing two wrong signs again has product $+1$. Hence one satisfied check assigns zero parity penalty to both the all-correct pattern and a pattern with two wrong signs. This is not a defect in the parity test: both local patterns have even parity. The channel evidence and overlapping checks must determine which local pattern is globally preferable.

Proposition 1 (Two-sign path through one check)**.** *Hold all but two members of one check at $+1$ and set the remaining two to $x_i = x_j = t$. The contribution of that check is*

$$J_{\text{check}}(t) = \frac{w_p}{n}(1-t^2)^2. \tag{5}$$

It is zero at $t = -1$ and $t = +1$, and it is positive for $|t| < 1$.

Proof. Along the declared path, the check product is t^2. Substitution into the second term of Equation (1) gives Equation (5). In particular, $J_{\text{check}}(0) = w_p/n > 0$. □

The contribution $(1-t^2)^2$ that favors hard magnitudes raises the same intervening hill. A gradient update can move every coordinate simultaneously; the issue is not that it literally flips one bit at a time. The issue is that even a collective small correction can initially increase the objective.

This proposition exhibits one uphill path for one check term. It does not prove that every continuous path between two codewords crosses a positive rim of the complete objective. Such a claim requires minimizing the maximum objective height over all connecting paths, together with a separately declared analysis of finite jumps.

4.1 A strict constructed local minimum on the box

The following example proves possibility without claiming frequency on the registered graph. On $[-1,1]^2$, define

$$\begin{aligned} F(x_1,x_2) = &-\frac{x_1+4.6x_2}{2}+2(1-x_1x_2)^2 \\ &+0.2\left[(1-x_1^2)^2+(1-x_2^2)^2\right]. \end{aligned} \tag{6}$$

The desired valid point has $F(+1,+1) = -2.8$. Nevertheless, the following holds.

Proposition 2 (Constructed wrong local minimum on the box)**.** *For the function in Equation* (6)*,*

$$x^* = (-1,-0.5), \qquad F(x^*) = 2.2625, \qquad \nabla F(x^*) = (0.5,0). \tag{7}$$

The point x^* is a strict local minimum on $[-1,1]^2$.

Proof. At the active lower bound $x_1 = -1$, any sufficiently small permitted change has $d_1 \geq 0$. Thus $\nabla F(x^*)^{\mathsf{T}} d = 0.5d_1 > 0$ whenever $d_1 > 0$. When $d_1 = 0$, the remaining change is along x_2, where the second derivative is

$$\frac{\partial^2 F}{\partial x_2^2}(x^*) = 3.8 > 0. \tag{8}$$

More strongly, the complete Hessian is

$$\nabla^2 F(x^*) = \begin{bmatrix} 2.6 & 0 \\ 0 & 3.8 \end{bmatrix} \succ 0. \tag{9}$$

By continuity, the Hessian remains positive definite in a sufficiently small neighborhood of x^*. Taylor expansion then gives a nonnegative linear term $0.5d_1$ for every permitted displacement and a strictly positive quadratic term for every nonzero displacement. Hence x^* is a strict local minimum on the box. □

For any $\eta > 0$,

$$\Pi_{[-1,1]^2}\left(x^* - \eta \nabla F(x^*)\right) = x^*. \tag{10}$$

The lower desired point exists, but an ordinary local projected-gradient step cannot reach it from x^*.

5 What Exact Projected Stationarity Proves

Let $\mathscr{C} = [-1,1]^n$. For a differentiable function J, define

$$G_\eta(x) = \frac{x - \Pi_{\mathscr{C}}\left(x - \eta \nabla J(x)\right)}{\eta}, \qquad \eta > 0. \tag{11}$$

Theorem 1 (Exact projected zero)**.** *For $x^* \in \mathscr{C}$ and $\eta > 0$,*

$$G_\eta(x^*) = 0 \quad \Longleftrightarrow \quad 0 \in \nabla J(x^*) + N_{\mathscr{C}}(x^*), \tag{12}$$

where $N_{\mathscr{C}}$ is the convex normal cone. Consequently, $\nabla J(x^*)^\mathsf{T} d \geq 0$ for every d such that $x^* + \epsilon d \in \mathscr{C}$ for all sufficiently small $\epsilon > 0$, and

$$\Pi_{\mathscr{C}}(x^* - \alpha \nabla J(x^*)) = x^* \qquad \text{for every } \alpha > 0. \tag{13}$$

Proof. The Euclidean projection characterization gives $x^* = \Pi_{\mathscr{C}}(x^* - \eta \nabla J(x^*))$ if and only if $-\eta \nabla J(x^*) \in N_{\mathscr{C}}(x^*)$. Because the normal cone is a cone, this is equivalent to Equation (12) and is unchanged by any positive scalar α. The final inequality follows directly from the definition of the normal cone. □

The frozen registered scalar protocol uses

$$D = \frac{13.6}{n} I, \qquad G_\eta^D(x) = \frac{x - \Pi_{\mathscr{C}}\left(x - \eta D^{-1} \nabla J_C(x)\right)}{\eta}. \tag{14}$$

Since D is a positive scalar multiple of the identity, this is the same Euclidean direction with a rescaled step. The exact zero sets coincide after the corresponding scalar conversion. Finite numerical tolerances do not: for $n = 10200$, D^{-1} scales an interior partial derivative by 750.

An exact projected zero therefore closes additional ordinary local $-\nabla J_C$ descent from that state. It does not close a route that avoids the basin, carries nonzero auxiliary velocity, takes a finite nonmonotone jump, changes the initialization, applies a nonscalar or state-dependent matrix before stationarity, changes the objective, or introduces new variables.

5.1 Registered and indicative evidence

The registered result is $0/200$ recovery at 1.5 dB through 90 updates for the frozen native compound control. It does not certify exact stationarity.

An external NumPy continuation on the same 200 exposed 1.5-dB frames reported $0/200$ recoveries, 197 terminal states meeting its own unscaled threshold $\|G_\eta(x)\|_\infty \leq 10^{-8}$, three 6000-update caps, a median of 744.5 wrong bits, and $J_C(x_{\text{word}}) < J_C(x_{\text{terminal}})$ on all 200 frames. This is useful indicative mechanism evidence, not a registered result. Its finite tolerance differs in scale from the registered D^{-1} diagnostic, and it did not perform the registered independent 10^{-10} confirmation pass.

The defensible interpretation is therefore that the external continuation is consistent with trajectories ending in wrong stationary regions. It does not certify local minimality. The registered result alone establishes only that the frozen method did not recover any of those 200 transmitted words within 90 updates.

6 A Convexity Boundary for Exact Parity Penalties

Parity can be placed in a differentiable cost. The obstruction is to asking one ordinary penalty with a finite coefficient to be simultaneously smooth, convex, exact, zero only at binary parity-valid points, and easy to minimize by ordinary gradient descent.

Let $\mathscr{V}_c$ be the set of binary even-parity patterns for check c, and let $\mathscr{P}_c = \text{conv}(\mathscr{V}_c)$ be its parity polytope [3].

Theorem 2 (Convex zero-set obstruction)**.** *Suppose p is nonnegative and convex and satisfies $p(v) = 0$ for every $v \in \mathscr{V}_c$. Then $p(u) = 0$ for every $u \in \mathscr{P}_c$.*

Proof. Every $u \in \mathscr{P}_c$ has a representation $u = \sum_{v \in \mathscr{V}_c} \alpha_v v$, where $\alpha_v \geq 0$ and $\sum_v \alpha_v = 1$. Convexity and nonnegativity give

$$0 \leq p(u) \leq \sum_{v \in \mathscr{V}_c} \alpha_v p(v) = 0. \tag{15}$$

Hence $p(u) = 0$ throughout $\mathscr{P}_c$. □

A convex penalty cannot therefore have precisely the multiple isolated binary even-parity patterns as its zero-valued global minima. The product construction obtains isolated hard zeros by giving up convexity, thereby permitting multiple local minima and intervening hills.

7 A Differentiable Convex Parity Cost

Let $b \in [0,1]^n$ be a shared vector, let $S_c b$ extract the entries incident on check c, and let f_{sig} be a differentiable signal cost. Consider

$$J_\rho(b) = f_{\text{sig}}(b) + \frac{\rho}{2} \sum_c \left\| S_c b - \Pi_{\mathscr{P}_c}(S_c b) \right\|^2. \tag{16}$$

Proposition 3 (Squared distance to the parity polytope)**.** *If every $\mathscr{P}_c$ is nonempty, closed, and convex, then the parity part of Equation* (16) *is convex and continuously differentiable. Moreover,*

$$\nabla J_\rho(b) = \nabla f_{\text{sig}}(b) + \rho \sum_c S_c^{\mathsf{T}} \left[S_c b - \Pi_{\mathscr{P}_c}(S_c b) \right]. \tag{17}$$

If f_{sig} is convex, then J_ρ is convex.

Proof. For a nonempty closed convex set P, the function $\frac{1}{2}\operatorname{dist}(u,P)^2$ is convex and continuously differentiable, with gradient $u - \Pi_P(u)$. Applying the chain rule to each linear extraction S_c yields Equation (17). Convexity is preserved by linear composition, nonnegative summation, and addition of f_{sig}. □

This parity contribution pulls each local configuration toward its parity polytope instead of creating a product hill between valid hard configurations. It does not guarantee a binary output: it is zero throughout each $\mathscr{P}_c$, including at fractional mixtures. Nor is the finite quadratic penalty generally exact.

The scalar example

$$\min_x x \quad \text{subject to} \quad x = 0 \tag{18}$$

has constrained solution $x = 0$. Replacing the constraint by

$$F_\rho(x) = x + \frac{\rho}{2} x^2 \tag{19}$$

gives $x_\rho = -1/\rho$, which violates the constraint for every finite ρ. Increasing ρ reduces the violation without making it exactly zero. In Equation (16), the restoring term in Equation (17) vanishes at feasibility, so a nonzero signal gradient can balance it at a slightly infeasible point.

Exposed-population diagnostic. Experiment `RQ2-CONVEX-PARITY-DISTANCE-001` evaluated one frozen instance of Equation (16) on the already exposed 1.5-dB frames 61969–62168. It retained one scalar $b_i \in [0,1]$ per bit, used the linear signal term with $\rho = 96$, calculated the partial derivatives through exact Euclidean projection onto every degree-six parity polytope, applied the fixed step $1/288$, and projected componentwise onto the box. Its frozen source defines no sigmoid, restart, momentum, multiplier, or state-dependent matrix, and its candidate packet contains no BP quantity.

The latched transmitted-word recovery count was $0/200$ at updates 30, 90, 300, 1000, 3000, and 6000. No parity-valid hard-decision estimate occurred at any update. No stored candidate state, including every checkpoint and terminal, had a readout tie. Every case reached the update-6000 cap with $\|G(b)\|_\infty > 10^{-8}$, so all 200 terminals are `optimization_unresolved`. The terminal mapping-norm median was $2.2772885202382298 \times 10^{-4}$, with range 2.4881×10^{-5} to 0.0115097. After the post-commit scorer opened the transmitted word, the terminal hard-decision estimates had median 382 coded-bit errors (range 148–595) and median syndrome weight 387 (range 224–508). The soft states had median 3274 fractional coordinates and median maximum local check distance 0.102299. Primary and replay scientific traces were identical.

Although the objective values were essentially flat late in the run, none of the cases met the frozen numerical-stationarity tolerance. Convexity therefore does not license a claim that these terminals are global minimizers, and the result does not close this objective. The source-only follow-up below tested faster same-objective convergence without clearing its population-run gate. The exact constrained replica-and-multiplier formulation and the untouched two-hop branch were not tested.

Source-only solver-successor gates. Production-size synthetic gates on the same frozen objective did not reach $\|G_{288}\|_\infty \le 10^{-8}$, so no second 1.5-dB population run was launched. The feasible accelerated arm ended after 6000 updates at $\|G_{288}\|_\infty = 1.8247366 \times 10^{-3}$; its synthetic suite passed 44,031/44,031 assertions. A hybrid of those 6000 updates and 30,000 monotone local-curvature updates ended at $\|G_{288}\|_\infty = 2.3504324 \times 10^{-4}$ after 73,511 evaluations and 43,510 backtracks. Optim.jl 2.3.2 `LBFGSB`, capped at 5000 iterations, ended at $\|G_{288}\|_\infty = 5.4017116 \times 10^{-5}$ for $m = 10$ and 6.3955257×10^{-5} for $m = 50$. The local-curvature suite passed 15,233/15,233 assertions.

These are synthetic source-feasibility diagnostics, not 1.5-dB population evidence and not evidence that the objective fails. A decisive global-minimum audit now requires a materially stronger certified optimizer, or a relaxed tolerance paired with an explicit error bound. The active two-hop branch is untouched.

8 Replica Variables and Multipliers

Exact consistency can instead remain a constraint. Introduce one replica variable z_c for each check and impose

$$z_c = S_c b, \qquad z_c \in \mathcal{P}_c. \tag{20}$$

With residual $r_c = S_c b - z_c$ and multiplier y_c, the augmented Lagrangian is

$$\mathcal{L}_\rho(b, z, y) = f_{\text{sig}}(b) + \sum_c \left[y_c^{\mathsf{T}} r_c + \frac{\rho}{2} \|r_c\|^2 \right], \qquad z_c \in \mathcal{P}_c. \tag{21}$$

Its partial derivative with respect to the shared vector is

$$\nabla_b \mathcal{L}_\rho = \nabla f_{\text{sig}}(b) + \sum_c S_c^{\mathsf{T}} (y_c + \rho r_c). \tag{22}$$

At exact consistency, $r_c = 0$, the quadratic contribution disappears but the multiplier need not. The first-order balance can remain

$$\nabla f_{\mathrm{sig}}(b) + \sum_c S_c^{\mathsf{T}} y_c = 0. \tag{23}$$

Thus the multipliers can retain a nonzero normal force while the shared and replica variables agree. A typical alternating direction method updates the shared variables, projects each replica variable onto its parity polytope, and updates the multipliers from the residuals [3], [4]. This uses partial derivatives, but it is not ordinary descent of the unconstrained scalar objective J_C. It changes the variables, feasible sets, and update operation.

9 Exactness Boundary on a Loopy Tanner Graph

Each $\mathcal{P}_c$ exactly describes the convex hull of the valid binary patterns for one parity check. On a loopy Tanner graph, however, intersecting these local polytopes through the shared variables gives the fundamental-polytope relaxation used in LP decoding [2]. That relaxation can have fractional vertices, commonly called pseudocodewords in the LP-decoding literature.

Consequently,

$$\begin{gathered}\text{exact feasibility in every check's parity polytope} \\ \text{does not guarantee a binary codeword.}\end{gathered} \tag{24}$$

A linear cost over the exact global codeword polytope has an optimal codeword vertex. A unique optimum selects that vertex. If several codewords tie, fractional mixtures on the optimal face are also optimal and require a deterministic vertex or tie rule. Describing or projecting onto the exact global polytope can be substantially more expensive than using the relaxation formed from the individual checks. The difficulty therefore moves from a nonconvex product landscape into the representation and enforcement of the feasible set; it does not disappear.

10 Prospective Research Direction

The mathematically distinct successor formulation is

$$\boxed{\begin{aligned}&\text{linear or differentiable lifted signal cost} \\ &\quad + \text{ parity-polytope feasibility} \\ &\quad + \text{ shared-to-local consistency and multipliers}\end{aligned}} \tag{25}$$

The finite differentiable penalty in Equation (16) now has the exposed-population diagnostic reported above, but that run stopped above its mapping tolerance and does not identify its global minimizers. The expanded constrained formulation in this section still has no recovery evidence in the present repository and is not introduced as a validated decoder. A prospective evaluation would need to declare its state dimension, projection algorithm, multiplier update, initialization, stopping and readout rules, deterministic replay checks, operation and memory budgets, and recovery and coded-bit-error metrics before opening held-out cases.

It also does not supersede the separate state-dependent two-hop matrix branch. That branch applies a bounded matrix to the partial derivatives of J_C and has a retained teacher-labelled fit, but no fresh direction-screen or decoder result. It must be completed or amended under its own registered gates.

11 Conclusion

The observed decoding failure need not mean that the calculated partial derivatives are wrong. The mathematics establishes a more specific possibility: a descent method can faithfully optimize an objective term structure that permits a wrong local minimum on the box. A satisfied check cannot distinguish the all-correct hard pattern from a hard pattern with two wrong signs, and the smooth product term can rise along a collective correction between them. The constructed example proves that this term structure can yield a strict wrong local minimum on the box even when the desired word has lower cost.

The experimental conclusion remains narrower. The frozen scalar protocol recovered $0/200$ words at 1.5 dB through 90 registered updates. The external continuation is consistent with trajectories ending in wrong stationary regions, but it is not a registered exact-stationarity certificate. The evidence closes the frozen scalar step grid and normalization against the specific budget extension from 30 to 90 updates. As a research-priority decision, further repetitions that preserve the same state, objective, direction, and projection while changing only scalar implementation details should be deprioritized. Arbitrary discrete scalar-step schedules remain mathematically unclosed, and the evidence does not justify abandoning every use of partial derivatives.

The exposed convex parity-distance diagnostic removed nonglobal local wells from the tested objective but recovered $0/200$ words through update 6000 and produced no parity-valid hard-decision estimate. This is not yet evidence that the convex objective has the wrong global minimizer: every case stopped at the cap above the frozen mapping tolerance. The source-only successor gates also remained above tolerance, so no second population run was launched. The immediate question is therefore convergence, not another interpretation of the unresolved terminals. A decisive global-minimum audit requires a materially stronger certified optimizer, or a relaxed tolerance paired with an explicit error bound; only then can the signal cost, finite penalty, and fundamental-polytope relaxation be classified.

The constrained formulation changes what differentiation is asked to do. A differentiable signal cost ranks candidate explanations; parity polytopes, projections, and multipliers enforce the coding structure. Whether that expanded constrained formulation can approach BP at both 6 dB and 1.5 dB remains a prospective experimental question.

Reproducibility and Evidence Boundary

This paper is derived from the repository's explanatory and analytical records. The implemented objective is in `../implementation/two_hop_metric_action/compound_objective_jc.jl`. The full projected-descent derivation and result classification are in `../research/jc_projected_gradient_boundary.md`. The registered but unfinished continuation card is `../research/rq2_jc_projected_stationarity_001_card.md`. The active two-hop branch is scoped in `../research/active_goal.md`. The convex parity-distance diagnostic is specified in `../research/rq2_convex_parity_distance_001_card.md`, implemented in `../implementation/convex_parity_distance/`, and committed under `../results/rq2_convex_parity_distance_001/2026-11-08_v1/`. Its candidate and replay scientific-trace SHA-256 values both equal `2020306bb9d09d20e025528c2b9eba8a398f67e8c896b7329c8b3411c1c1dd20`. It used only the already exposed frames 61969–62168; the candidate packet contained no BP quantity, the frozen candidate source contained no BP call, and the run did not open the untouched confirmation ranges.

References

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